



Kadir Has University – Computer Engineering Department

CCIP503 Computer Networks & Security

Week – 2

Computer Networks and the Internet

**Assist. Prof. Dr.
İlktan AR**





0.1 Course – Tentative Outline

Week	Date	Topic
1	28 Sept.	Computer Networks and the Internet
2	5 Oct.	Computer Networks and the Internet
3	12 Oct.	Application Layer
4	26 Oct.	Application Layer
5	2 Nov.	Transport Layer, Wireshark usage and examples
6	9 Nov.	First Midterm Exam
7	16 Nov.	Transport Layer
8	23 Nov.	The Network Layer: Data Plane
9	30 Nov.	The Network Layer: Control Plane
10	30 Nov.	The Link Layer and Local Area Networks
11	7 Dec.	Second Midterm Exam
12	14 Dec.	Security Technology and Principles
13	21 Dec.	Security Technology and Principles
14	28 Dec.	Security Technology and Principles, Talk for Final Exam
FN	???	Final Exam

Updated!

Updated!

0.2 Previous Week

- We will focus on chapters 1.4, 1.5, 1.6, 1.7.

Your duties:

- ✓ Review this lecture note via learn.khas.edu.tr
- ✓ Find and download textbook.
- ✓ Read chapter 1 from Kurose & Ross's textbook.
- ✓ Download and Install WireShark application.





Any Questions



Week 2 - Agenda

01

Delay, Loss and Throughput

02

Protocol Layers and Service Models

03

Networks Under Attack

04

History of Networking and Internet

05

Next Week

1.1 Introduction

- Computer networks necessarily constrain throughput (the amount of data per second that can be transferred) between end systems, introduce delays between end systems and can lose packets.
- The physical laws of reality introduce delay and loss as well as constrain throughput.



1.2 Introduction

- As a packet travels from one node (host or router) to the subsequent node (host or router) along this path, the packet suffers from several types of delays at *each* node along the path.
- The most important of these delays are the **nodal processing delay**, **queuing delay**, **transmission delay**, and **propagation delay**; together, these delays accumulate to give a **total nodal delay**.

Nodal Tr: Düğümset

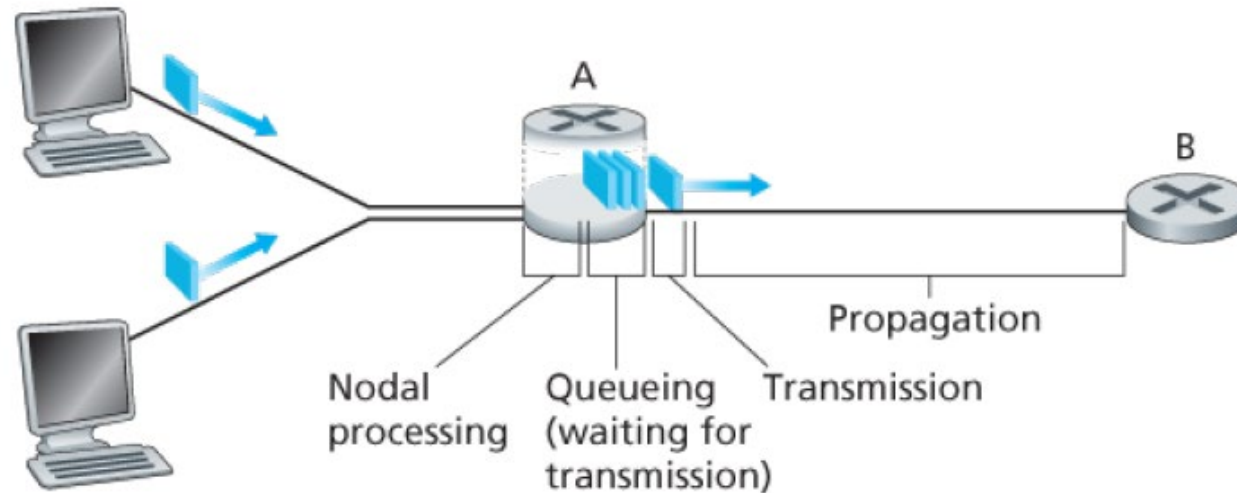




1.3 What is Delay?

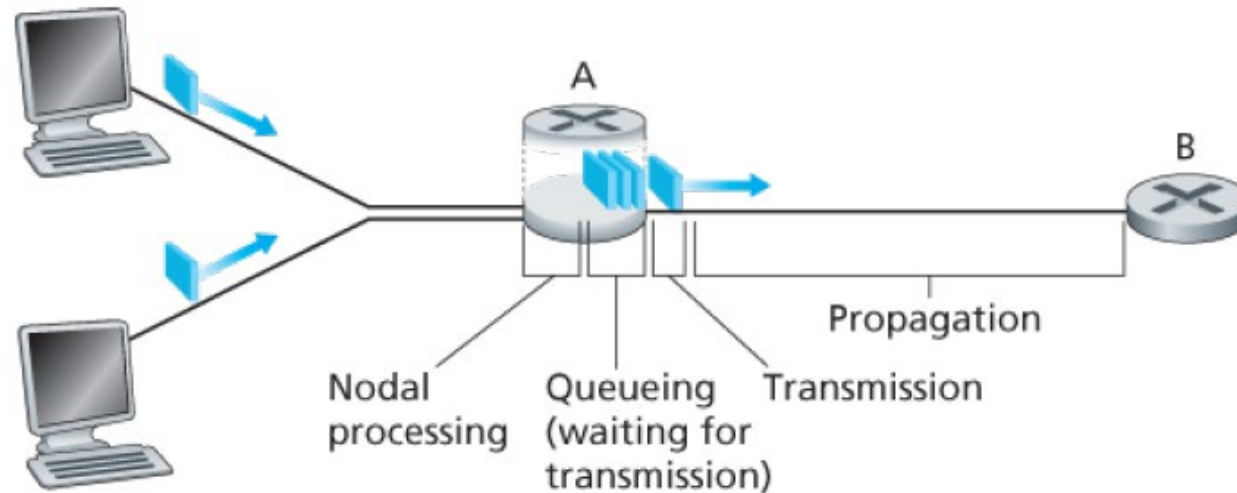
- **Delay:** Total time taken for a packet to travel from source to destination.
- Components of delay:
 - Processing Delay
 - Queuing Delay
 - Transmission Delay
 - Propagation Delay
- **End-to-End Delay:** Sum of all delays experienced by a packet.

1.4 Types of Delay



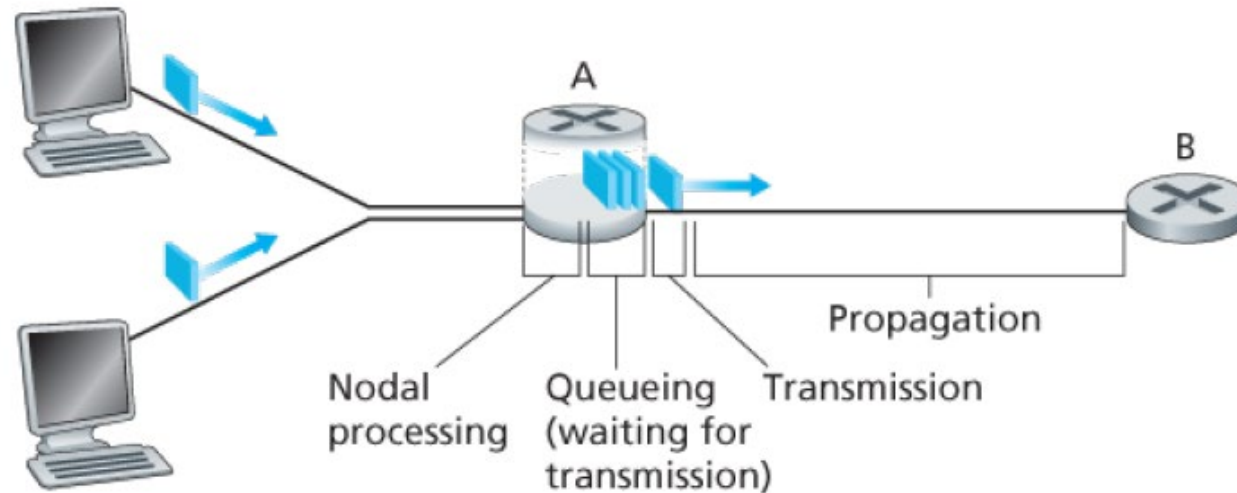
- The time required to examine the packet's header and determine where to direct the packet is part of the **(nodal) processing delay**.
- The processing delay can also include other factors, such as the time needed to check for bit-level errors in the packet that occurred in transmitting the packet's bits from the upstream node to router A.
- Processing delays in high-speed routers are typically on the order of microseconds (10^{-6} second) or less.

1.5 Types of Delay (cont.)



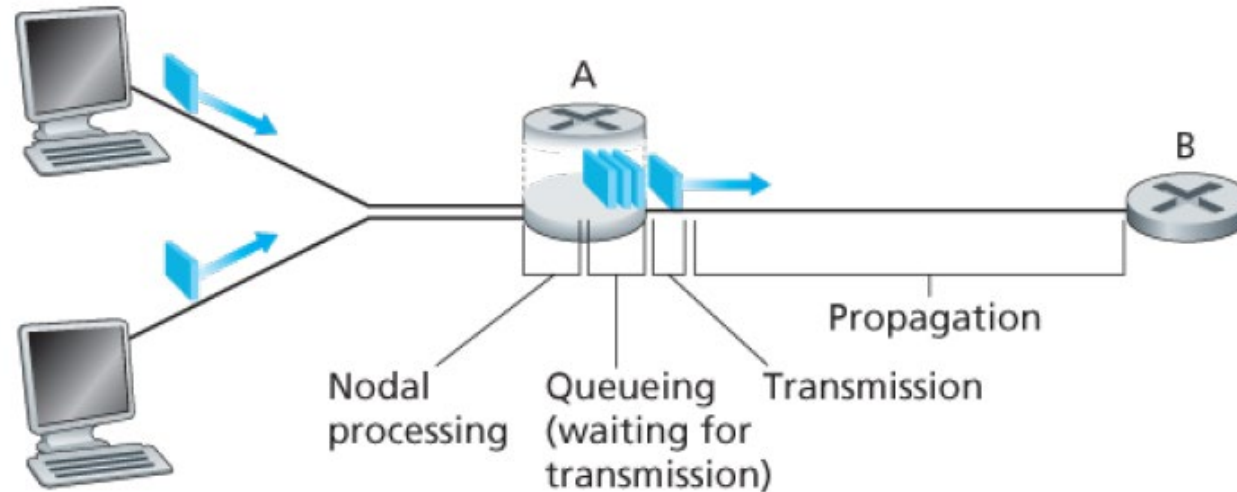
- At the queue, the packet experiences a **queuing delay** as it waits to be transmitted onto the link.
- The length of the queuing delay of a specific packet will depend on the number of earlier-arriving packets that are queued and waiting for transmission onto the link.
- Queuing delays can be on the order of microseconds to milliseconds (10^{-3} second) in practice.

1.6 Types of Delay (cont.)



- Assuming that packets are transmitted in a first-come-first-served manner and the length of the packet by L bits, and denote the transmission rate of the link from router A to router B by R bits/sec.
- The **transmission delay** is L/R .
- This is the amount of time required to push (that is, transmit) all the packet's bits into the link.
- Transmission delays are typically on the order of microseconds to milliseconds in practice.
- It can be hundreds of milliseconds for large Internet packets sent over low-speed dial-up modem links.

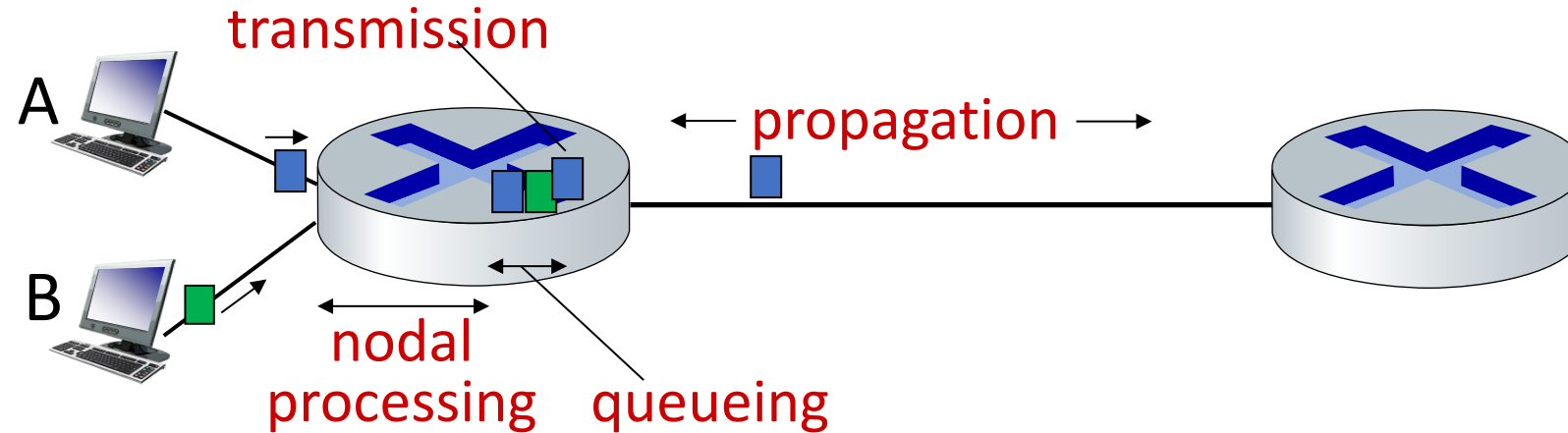
1.7 Types of Delay (cont.)



- Once a bit is pushed into the link, it needs to propagate to router B. The time required to propagate from the beginning of the link to router B is the **propagation delay**.
- The bit propagates at the propagation speed of the link. The propagation speed depends on the physical medium of the link (that is, fiber optics, twisted-pair copper wire, and so on).
- Prop. delay is hundreds of milliseconds for two routers interconnected by a geostationary (tr: sabit konumlu) satellite link.

Propagation Tr: yayılma

1.8 Types of Delay (All Together)



$$d = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

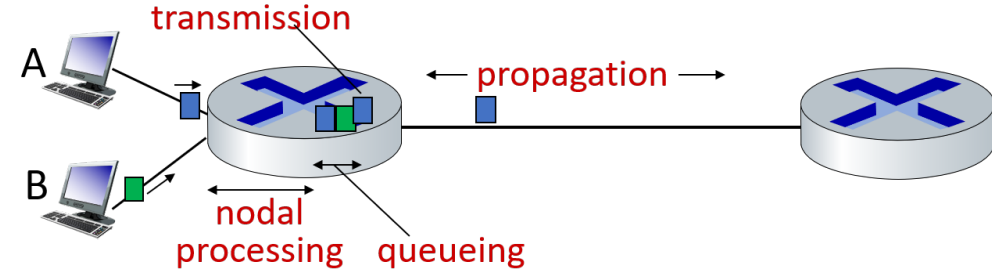
d_{trans} : transmission delay

- L : packet length (bits)
- R : link *transmission rate* (bps)
- $d_{\text{trans}} = L/R$

d_{prop} : propagation delay

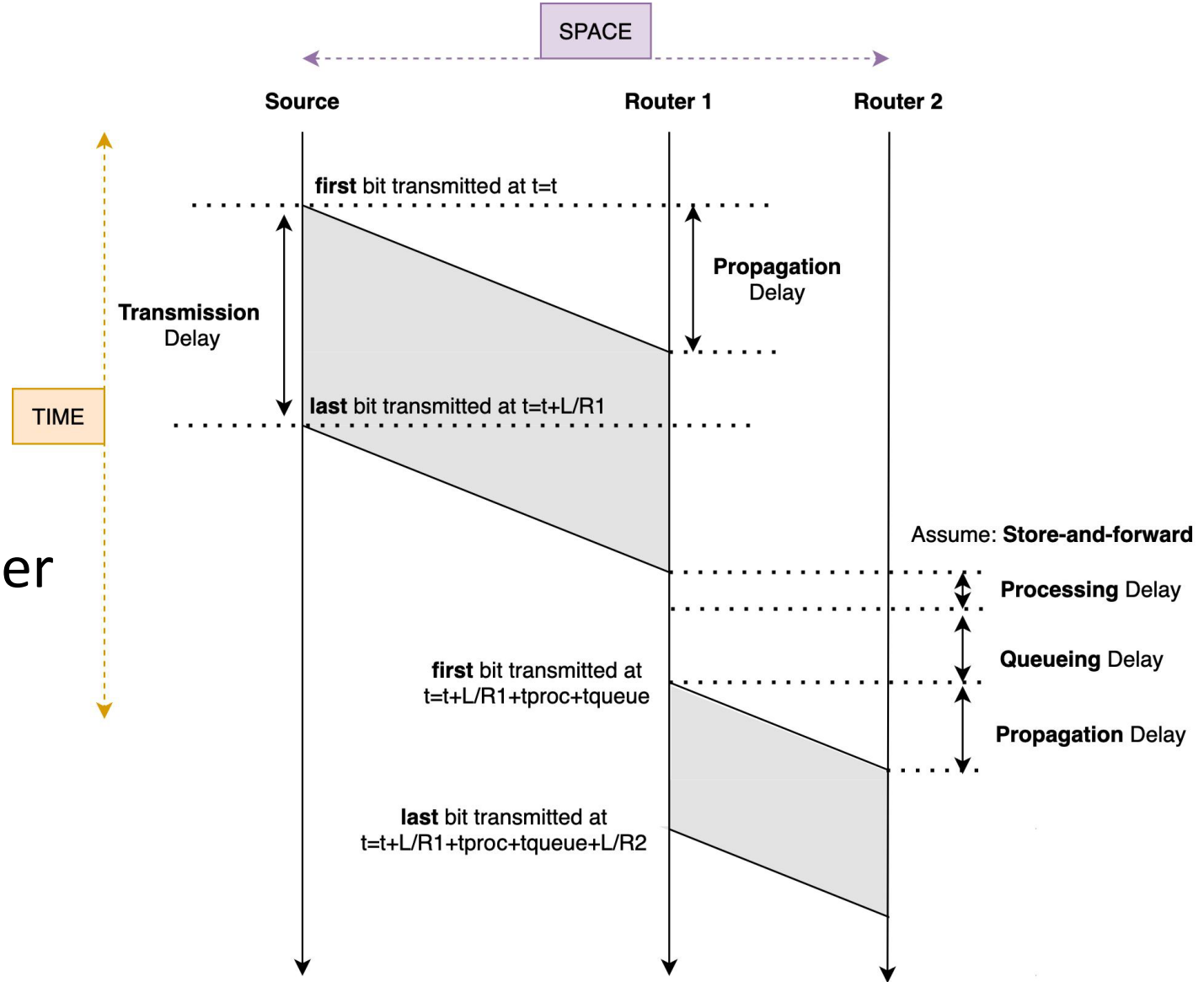
- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

1.9 Types of Delay (Example)



$$d = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- Assume link btw A and Left Router (*Router1*) has a capacity of $R1$.
- Assume link btw Routers has a capacity of $R2$.





1.10 Prediction Question: Which Delay Changes?

- Scenario A: packet size doubles, while link rate and distance stay unchanged.

d_{trans} transmission delay changes (increases)

- Scenario B: link distance doubles, while packet size and link rate stay unchanged.

d_{prop} propagation delay (increases)

- Scenario C: link rate doubles, while packet size and distance stay unchanged.

d_{trans} transmission delay changes (decreases)

1.11 Queuing Delay

- If 10 packets arrive at an empty queue at the same time, the first packet transmitted will suffer no queuing delay, while the last packet transmitted will suffer a relatively large queuing delay (while it waits for the other nine packets to be transmitted).
- Therefore, when characterizing queuing delay, one typically uses statistical measures, such as average queuing delay, variance of queuing delay, and the probability that the queuing delay exceeds some specified value.



1.12 Queueing Delay: A Simple Timeline

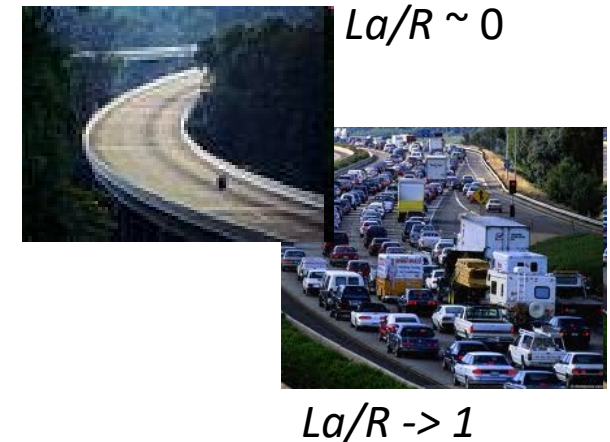
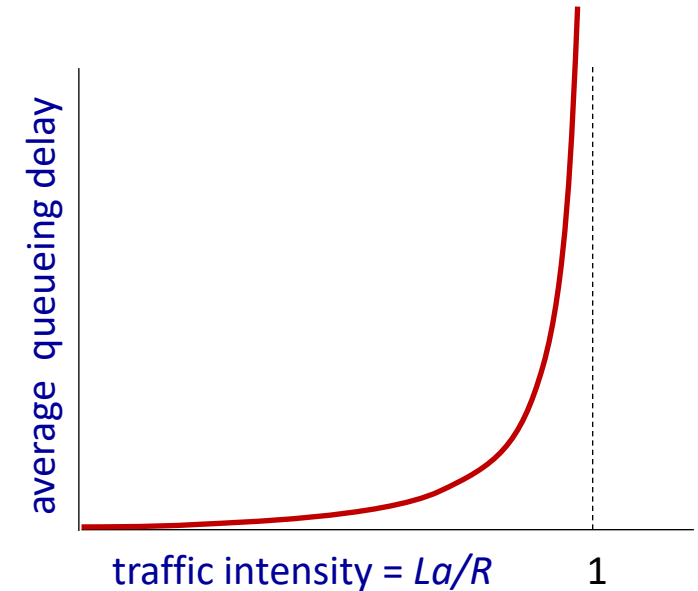
- Suppose one packet is being transmitted, and several packets are waiting behind it.
- The next packet cannot start transmission until the link becomes available.
- The waiting time is queueing delay.
- If packets arrive faster than the link can serve them for a sustained period, the queue grows.
- If the finite buffer becomes full, newly arriving packets can be dropped.

1.13 Queuing Delay (cont.)

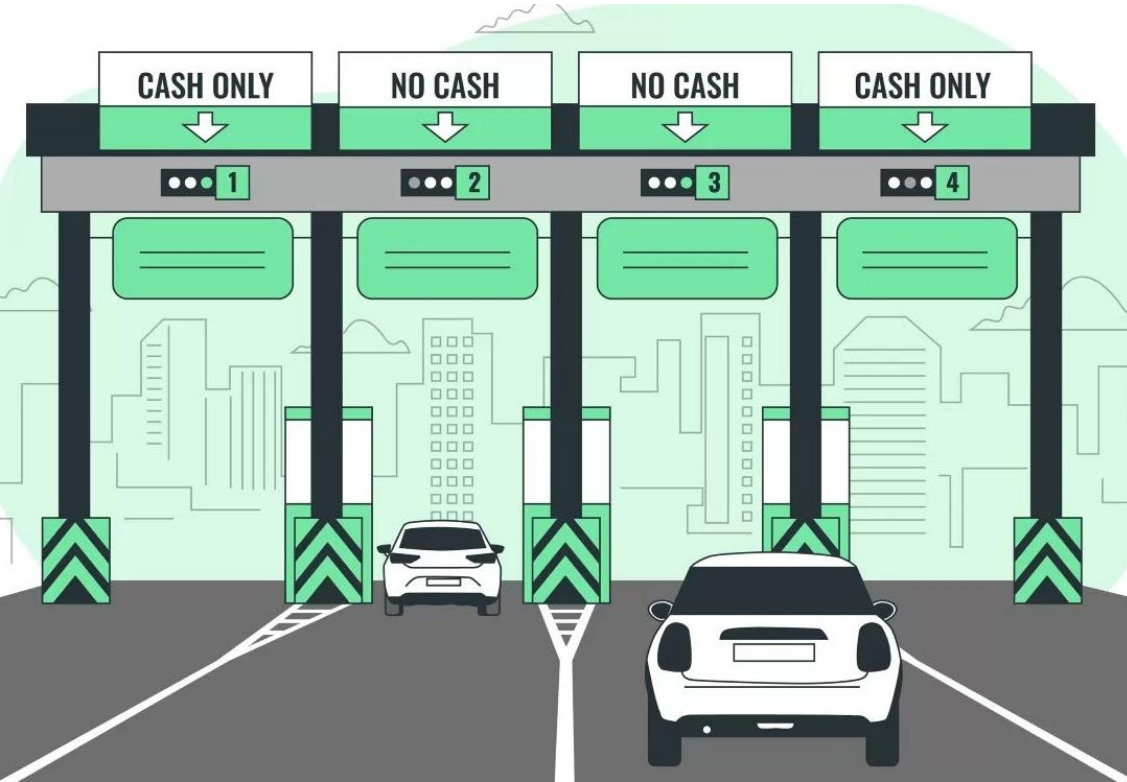
- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$

- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!



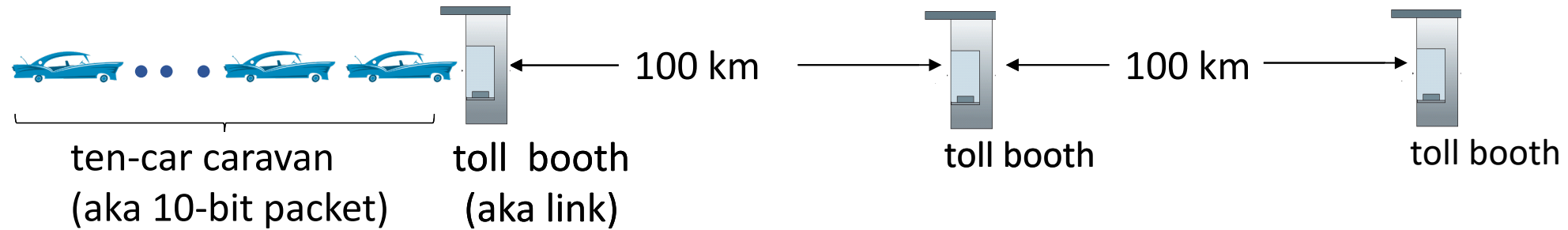
1.14 Queuing Delay (Example)



■ $L\lambda/R \sim 0$: avg. queueing delay small

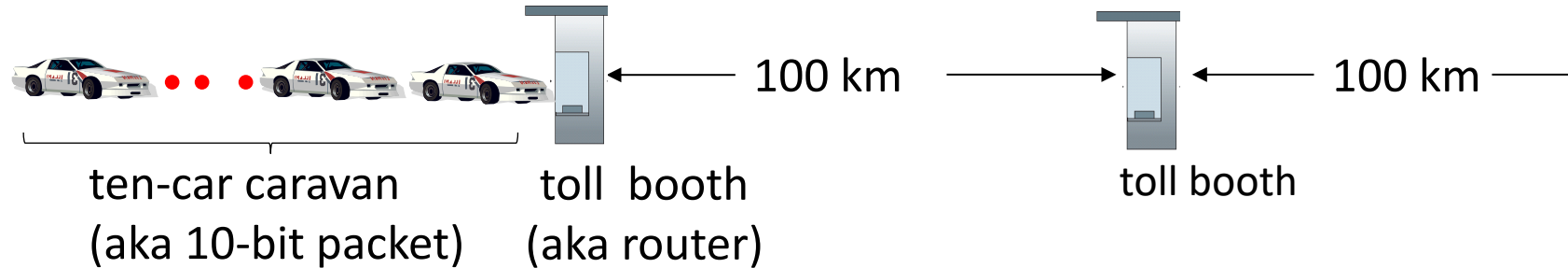
■ $L\lambda/R \rightarrow 1$: avg. queueing delay large

1.15 Caravan Analogy



- car $\sim$ bit; caravan $\sim$ packet; toll service $\sim$ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway = $12 * 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**

1.16 Caravan Analogy (cont.)



- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**
A: Yes! after 7 min,
first car arrives at second booth; three cars still at first booth.



1.17 Example: Calculating Delays

- Given:
 - Packet Length: 1,500 bytes
 - Link Bandwidth: 10 Mbps
 - Propagation Speed: 2×10^8 m/s
 - Link Length: 2,000 km
- Find:
 - 1. Transmission Delay
 - 2. Propagation Delay
- Transmission Delay = $L / R = (1500 \times 8) / (10 \times 10^6) = 1.2$ ms
- Propagation Delay = $d / s = 2000 \times 10^3 / (2 \times 10^8) = 10$ ms

1.18 Basic Problem: Transmission + Propagation

• PROBLEM

- Packet length = 2,000 bytes
- Link rate $R = 8$ Mbps
- Link length $d = 500$ km
- Propagation speed $s = 2 \times 10^8$ m/s
- Ignore processing and queuing delay.
- Find d_{trans} , d_{prop} , and total delay.

SOLUTION

$$L = 2,000 \times 8 = 16,000 \text{ bits.}$$

$$d_{\text{trans}} = L/R = 16,000/(8 \times 10^6) \text{ s} = 0.002 \text{ s} = 2 \text{ ms.}$$

$$d_{\text{prop}} = d/s = (500 \times 10^3)/(2 \times 10^8) \text{ s} = 0.0025 \text{ s} = 2.5 \text{ ms.}$$

$$\text{Total} = 2 \text{ ms} + 2.5 \text{ ms} = 4.5 \text{ ms.}$$

ANSWER: $d_{\text{trans}} = 2$ ms; $d_{\text{prop}} = 2.5$ ms; total = 4.5 ms



1.19 Intermediate Problem: Traffic Intensity

PROBLEM

- Average packet arrival rate $a = 500$ packets/s
- Average packet length $L = 1,000$ bytes
- Link rate $R = 5$ Mbps
- Compute traffic intensity $\rho = La/R$.
- Interpret the result.

SOLUTION

$$L = 1,000 \times 8 = 8,000 \text{ bits/packet.}$$

$$La = 8,000 \times 500 = 4,000,000 \text{ bits/s} = 4 \text{ Mbps.}$$

$$\rho = 4 \text{ Mbps} / 5 \text{ Mbps} = 0.8.$$

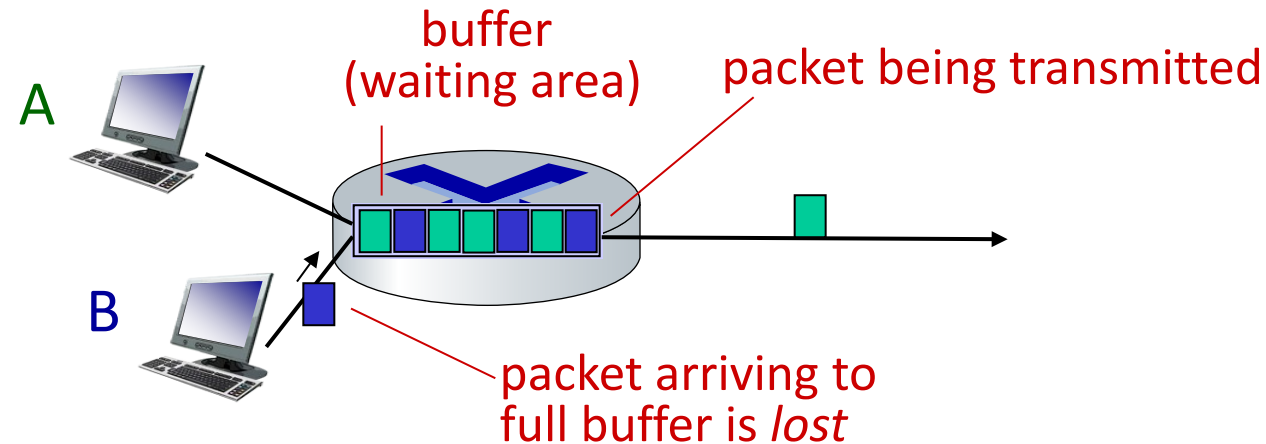
The offered workload is 80% of the link service capacity.

This is below 1, but the queue can still experience significant delay as ρ approaches 1.

ANSWER: $\rho = 0.80$ (80%)

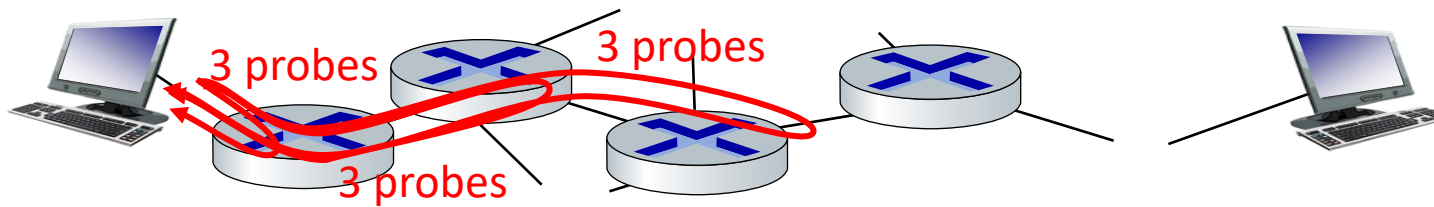
1.20 Packet Loss

- Queue (aka buffer) preceding link in buffer has finite capacity
- Packet arriving to full queue dropped (aka lost)
- Lost packet may be retransmitted by previous node, by source end system, or not at all



1.21 “Real” Internet Delays and Routes

- What do “real” Internet delay & loss look like?
- **traceroute** program provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply





1.22 End-to-End Delay

- Our discussion up to this point has focused on the nodal delay, that is, the delay at a single router.
- Let's now consider the total delay from source to destination.
- To get a hands-on feel for end-to-end delay in a computer network, we can make use of the traceroute program.
- The route was being traced from the source host **gaia.cs.umass.edu** (at the University of Massachusetts) to the **www.eurecom.fr** (University in France).



1.23 End-to-End Delay (traceroute)

- The output has six columns:
 - the first column is the n value which is the number of the router along the route;
 - the second column is the name of the router;
 - the third column is the address of the router (of the form xxx.xxx.xxx.xxx);
 - the last three columns are the round-trip delays for three experiments.
- If the source receives fewer than three messages from any given router (due to packet loss in the network), Traceroute places an asterisk just after the router number and reports fewer than three round-trip times for that router.



1.24 End-to-End Delay (Example traceroute)

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms	
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms	3 delay measurements to border1-rt-fa5-1-0.gw.umass.edu
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms	
4	jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms	
5	jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms	
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms	
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms	trans-oceanic link
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms	
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms	
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms	
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms	looks like delays decrease! Why?
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms	
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms	
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms	
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms	
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms	
17	***				
18	***				* means no response (probe lost, router not replying)
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms	

* Perform some traceroutes from exotic countries at www.traceroute.org



1.25 Throughput in Computer Networks

- In addition to delay and packet loss, another critical performance measure in computer networks is **end-to-end throughput**.
- To define throughput, consider transferring a large file from Host A to Host B across a computer network. This transfer might be, for example, a large video clip from one peer to another in a P2P file sharing system. The **instantaneous throughput** at any instant of time is the rate (in bits/sec) at which Host B is receiving the file.
- If the file consists of F bits and the transfer takes T seconds for Host B to receive all F bits, then the **average throughput** of the file transfer is F/T bits/sec.

Instantaneous tr: Anlık



1.26 Example: Calculating Throughput

- Scenario:
 - Server sends 10 MB file to client over 10 Mbps link.
 - Ignore loss and retransmissions.
- Find Throughput !
- Time = File Size / Link Rate = $(10 \times 8 \times 10^6) / (10 \times 10^6) = 8 \text{ s}$
- Throughput = 10 MB / 8 s = **1.25 MB/s**
- Throughput = Rate at which bits are successfully delivered from source to destination.



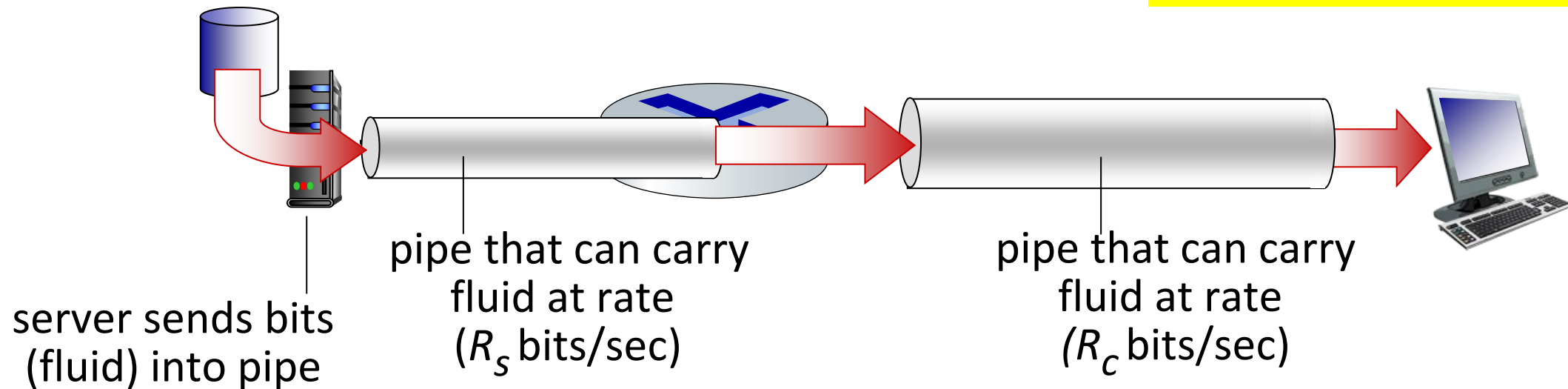
1.27 Delay-Throughput Trade-off

- **High Throughput**: Desirable, but may cause congestion (Buffer overflow) if rate $>$ network capacity.
- **Low Delay**: Also desirable, but buffering for high throughput can increase queuing delay.
- **Goal: Optimize both.**

1.28 Throughput in Computer Networks

- **Throughput:** rate (bits/time unit) at which bits are being sent from sender to receiver
 - **Instantaneous:** rate at given point in time
 - **Average:** rate over longer period of time

S for server, C for client, R for rate of link



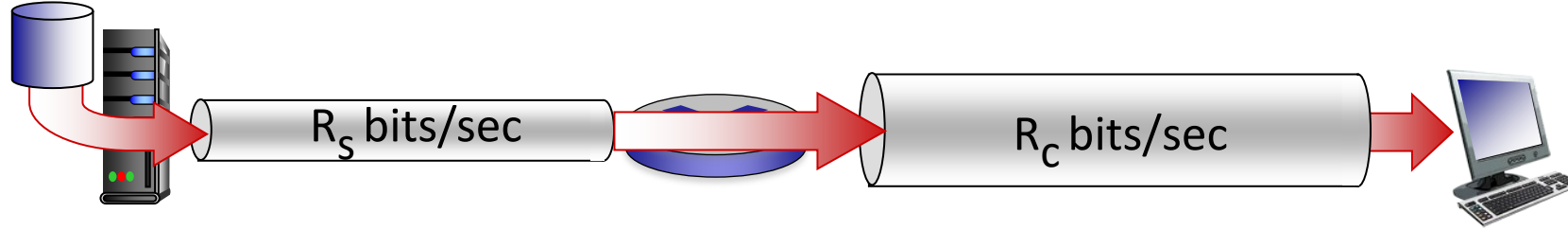


1.29 Throughput: Three Rates to Compare

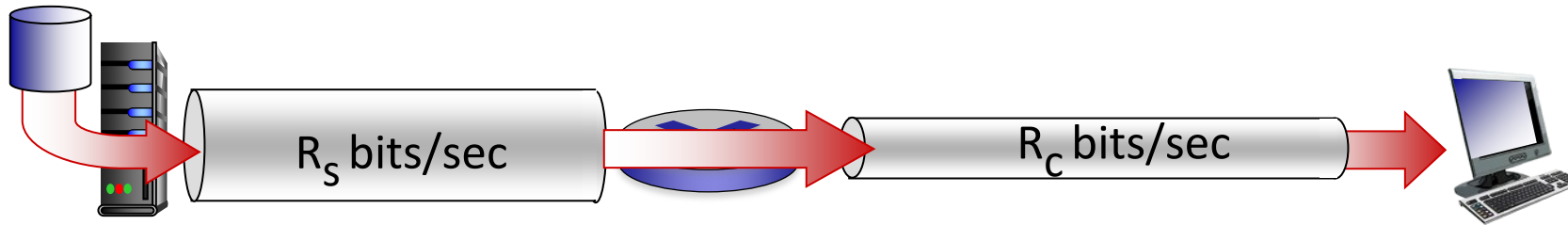
- When analyzing an end-to-end transfer, compare the sender rate, receiver-side capacity, and intermediate link capacities.
- The end-to-end throughput cannot exceed the capacity of the limiting part of the path.
- This is the intuition behind the bottleneck link.
- Important distinction: link capacity is a potential rate; measured throughput is the rate achieved by the transfer.

1.30 Bottleneck Link

$$R_s < R_c$$



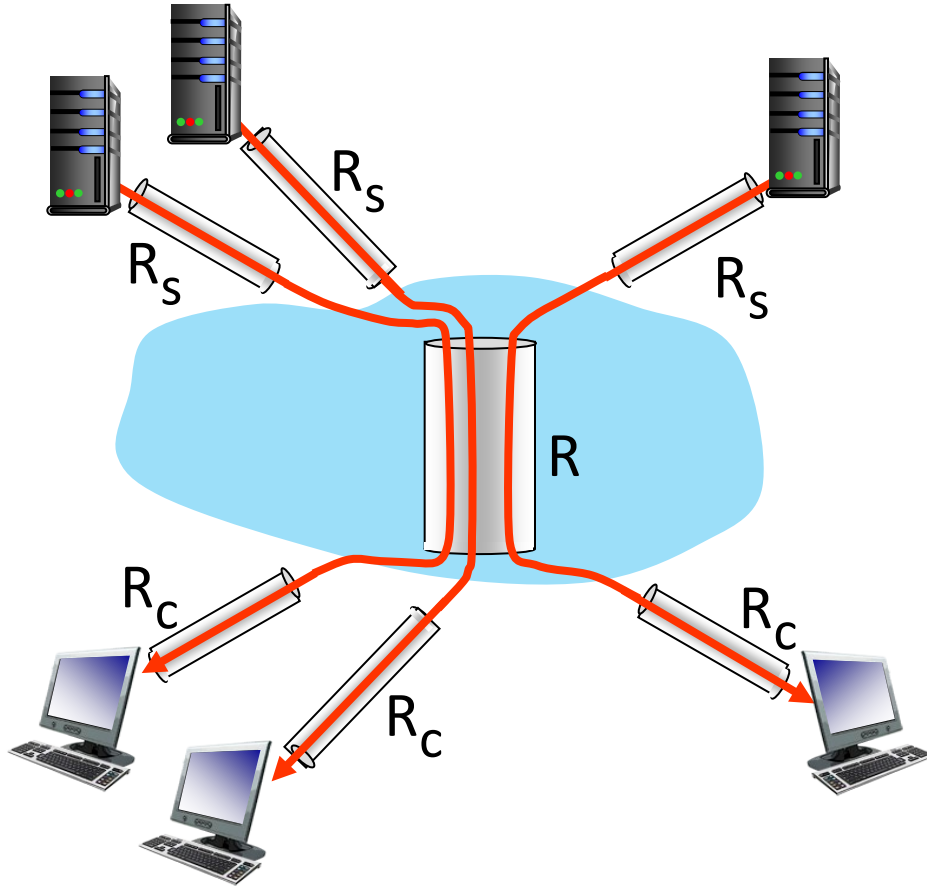
$$R_s > R_c$$



bottleneck link

link on end-end path that constrains end-end throughput

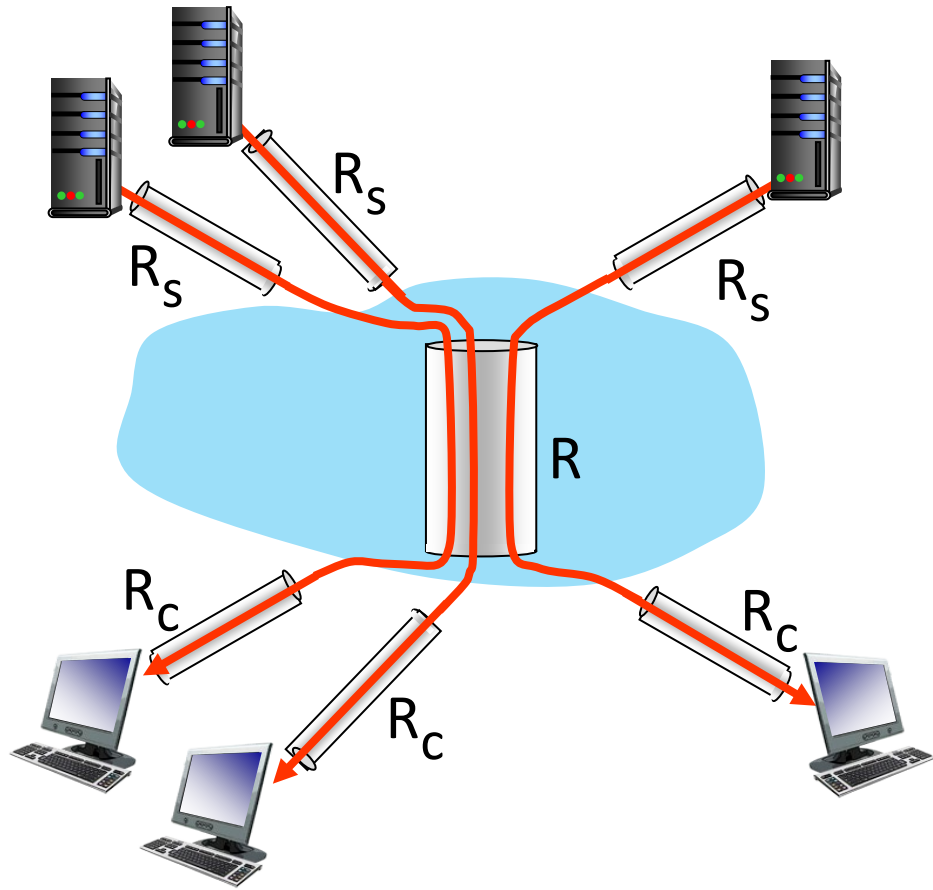
1.31 Bottleneck Link Example



10 connections (fairly) share
backbone bottleneck link R bits/sec

- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

1.32 Problem: Bottleneck Link



10 connections (fairly) share
backbone bottleneck link R bits/sec

PROBLEM

Server-side rate $R_s = 50$ Mbps.

Client-side rate $R_c = 20$ Mbps.

Backbone bottleneck link $R = 100$ Mbps.

Ten connections share R fairly.

Find the per-connection end-to-end throughput.

SOLUTION

Fair share of backbone link = $100/10 = 10$ Mbps.

Per-connection throughput = $\min(R_s, R_c, R/10)$.

Therefore: $\min(50, 20, 10)$ Mbps = 10 Mbps.

ANSWER: 10 Mbps per connection

A vibrant, abstract background consisting of numerous colorful paint splashes in shades of red, yellow, blue, and green, radiating from the center.

Any Questions

**10 mins break,
QR attendance!**

**TIME
FOR
BREAK!**



Week 2 - Agenda

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Delay, Loss and Throughput

02

Protocol Layers and Service Models

03

Networks Under Attack

04

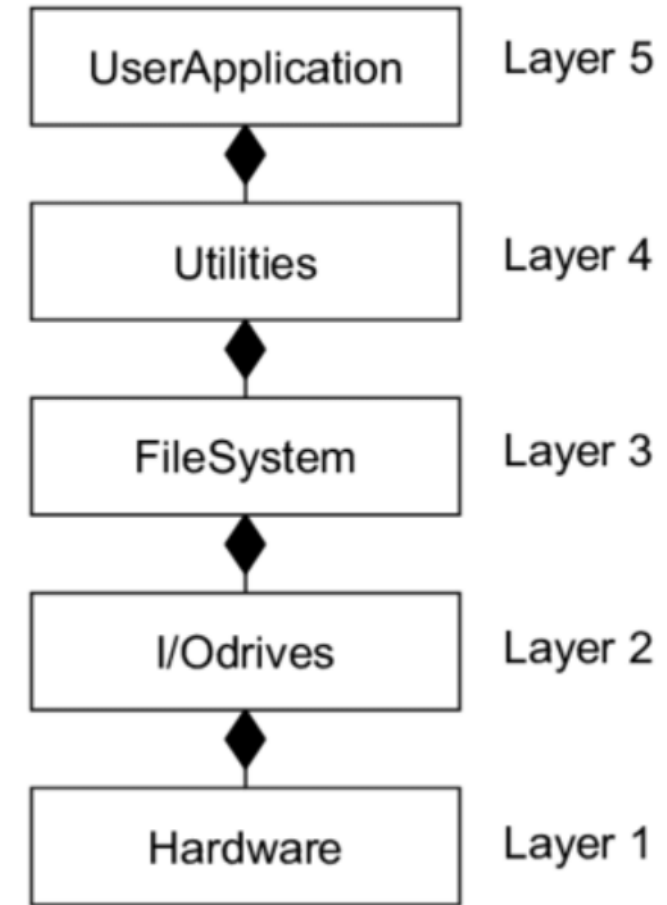
History of Networking and Internet

05

Next Week

2.1 Layered Architecture

- A layered architecture allows us to discuss a well-defined, specific part of a large and complex system.
- This simplification itself is of considerable value by providing modularity, making it much easier to change the implementation of the service provided by the layer.
- If the layer provides the same service to the layer above it, and uses the same services from the layer below it, the remainder of the system remains unchanged when a layer's implementation is changed.
- For large and complex systems that are constantly being updated, the ability to change the implementation of a service without affecting other components of the system is another important advantage of layering.



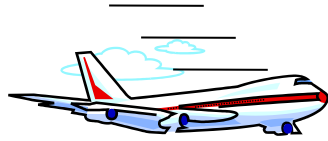
A simple layered OS architecture...



2.2 Protocol “layers” and Reference Models

- Networks are complex, with many “pieces”:
 - hosts
 - routers
 - links of various media
 - applications
 - protocols
 - hardware, software
- Question: is there any hope of organizing structure of network?

2.3 Example: Organization of Air Travel



————— *end-to-end transfer of person plus baggage* —————→

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

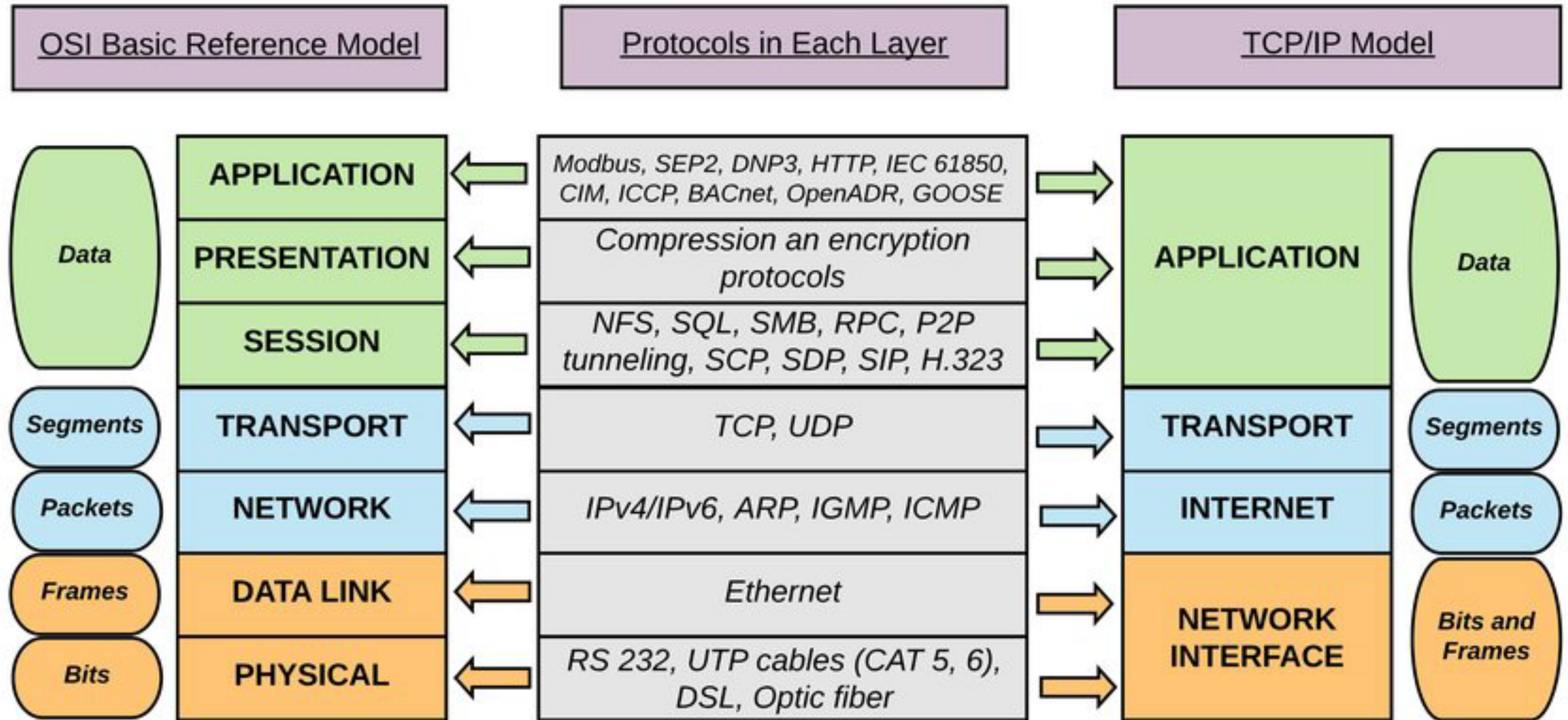
2.4 Example: Organization of Air Travel (cont.)



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

2.5 OSI Model and TCP/IP Model

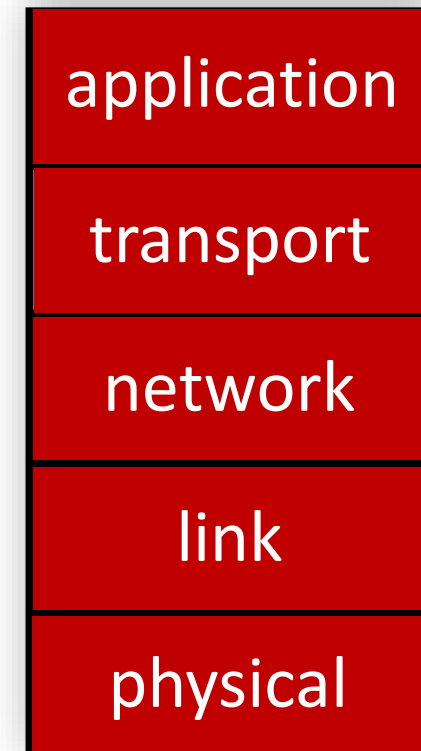


OSI: Open Systems Interconnection

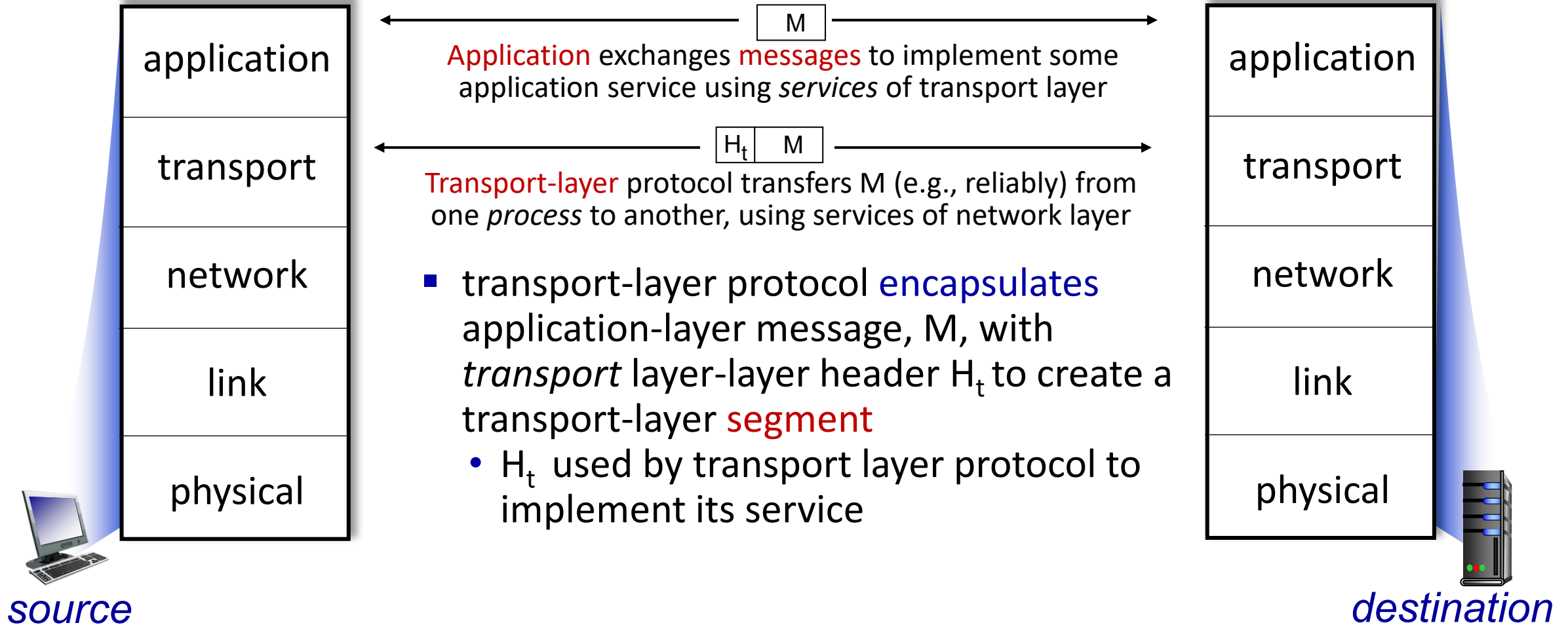
2.6 Layered Internet Protocol Stack (textbook)

The protocols of the various layers are called the **protocol stack**

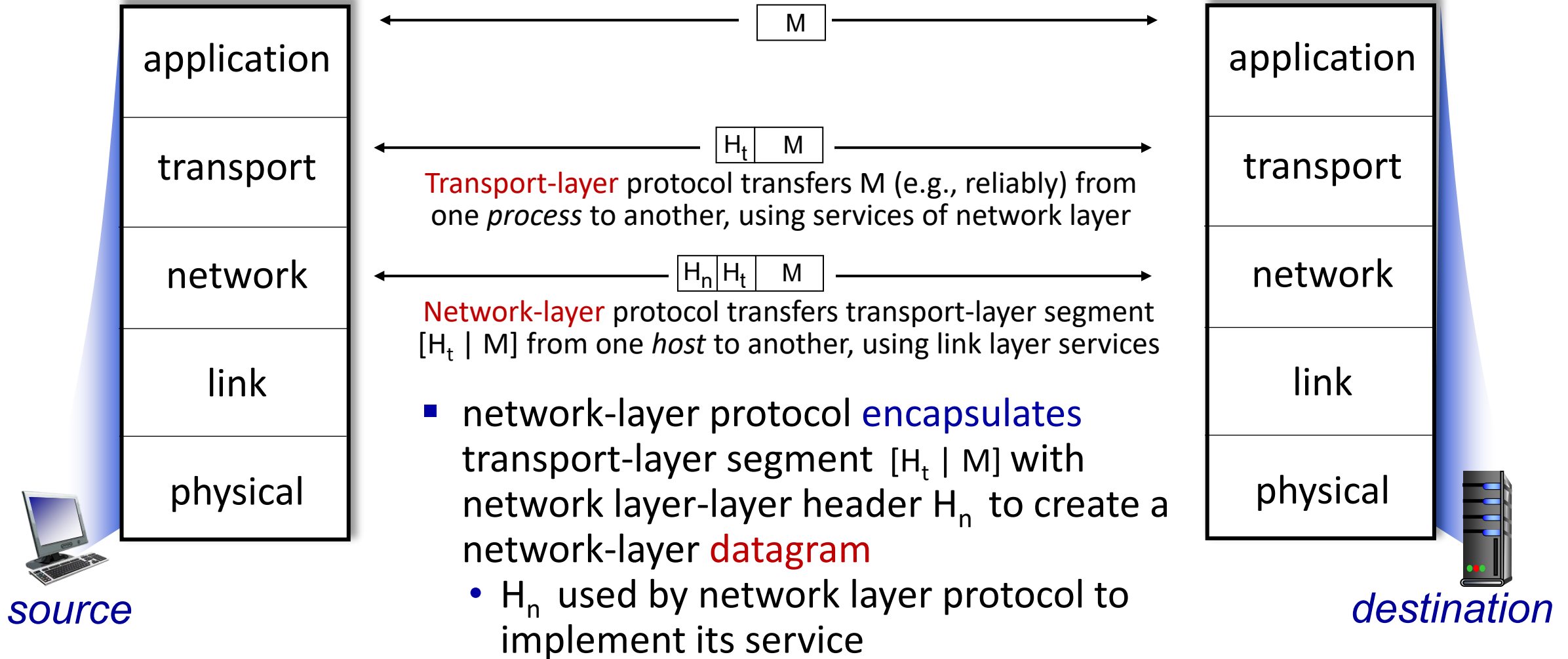
- **Application:** supporting network applications
 - HTTP, IMAP, SMTP, DNS
- **Transport:** process-process data transfer
 - TCP, UDP
- **Network:** routing of datagrams from source to destination
 - IP, routing protocols
- **Link:** data transfer between neighboring network elements
 - Ethernet, 802.11 (WiFi), PPP
- **Physical:** bits “on the wire”



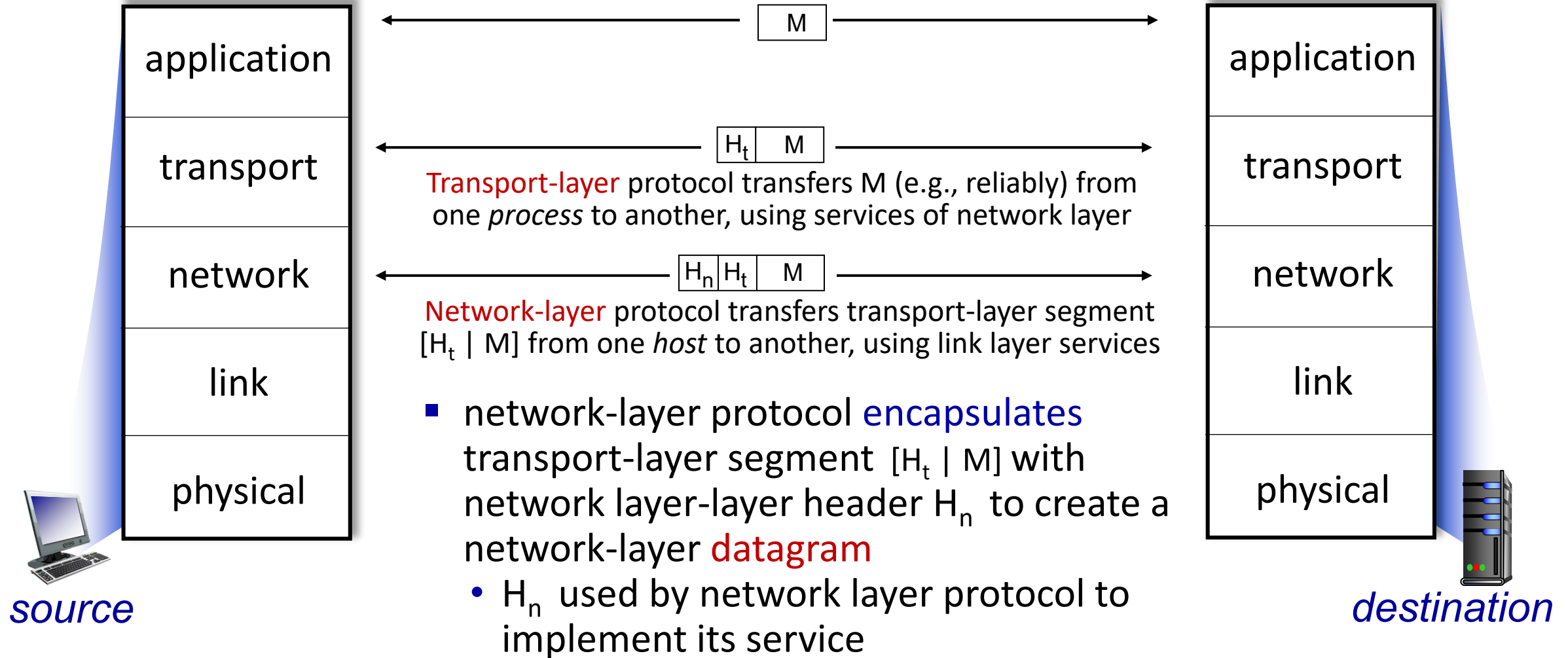
2.7 Services, Layering and Encapsulation



2.8 Services, Layering and Encapsulation

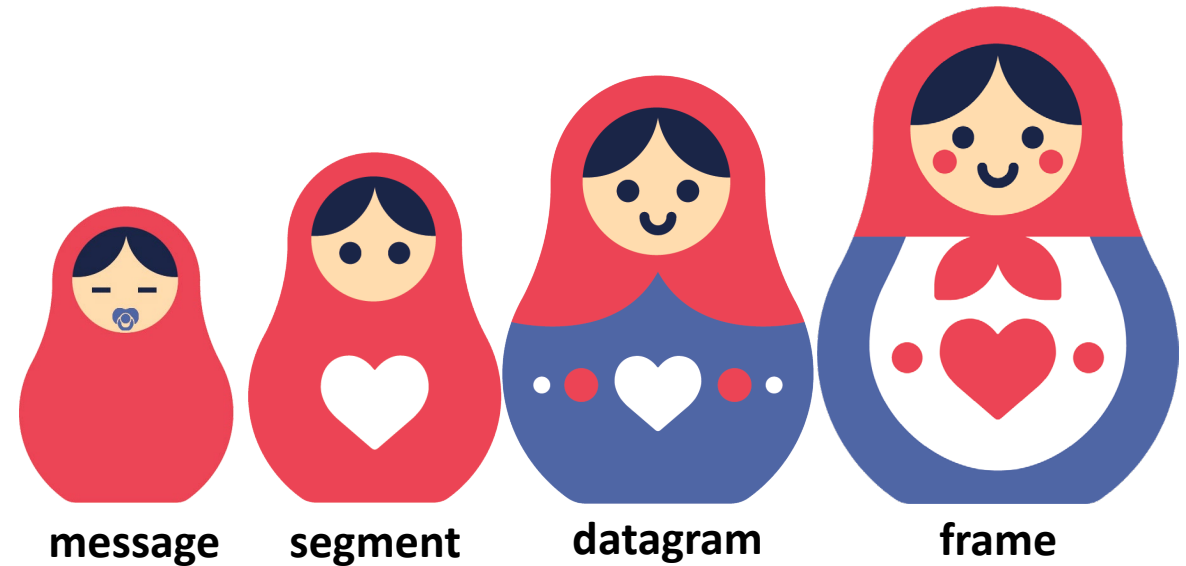


2.9 Services, Layering and Encapsulation

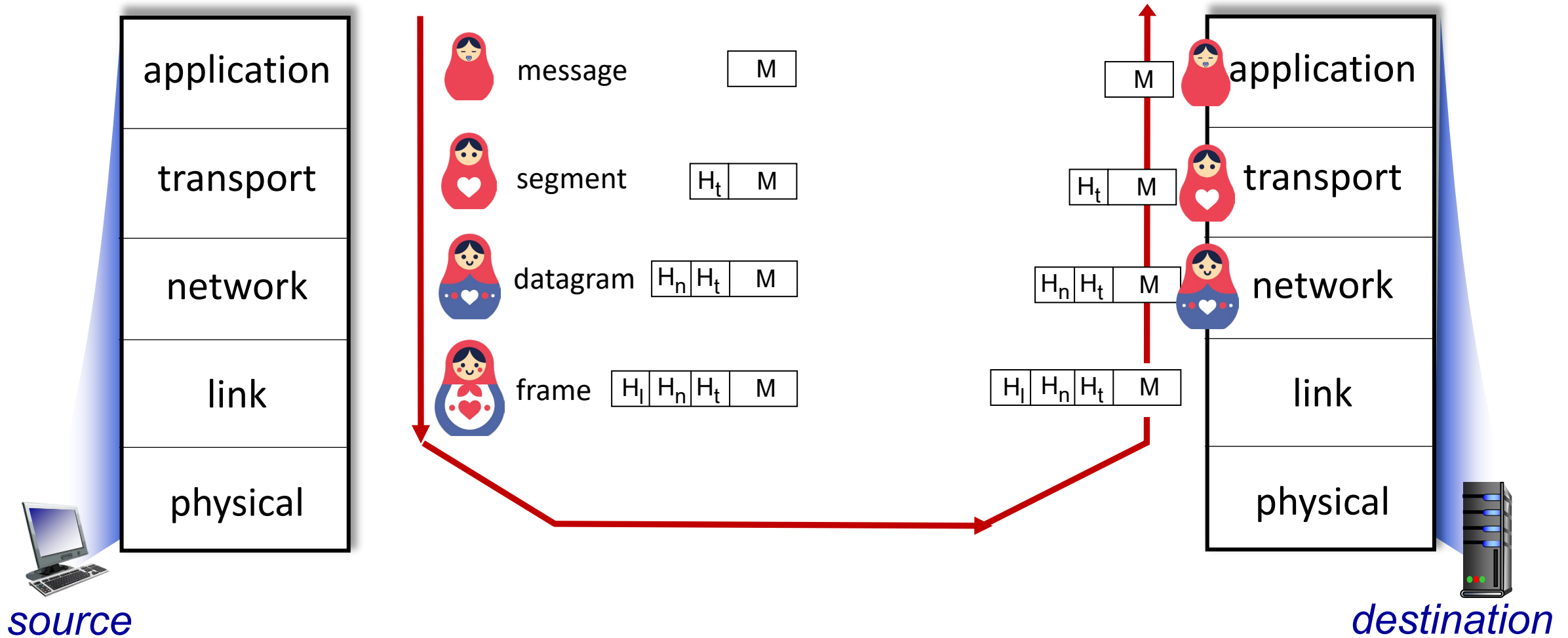


2.10 Encapsulation

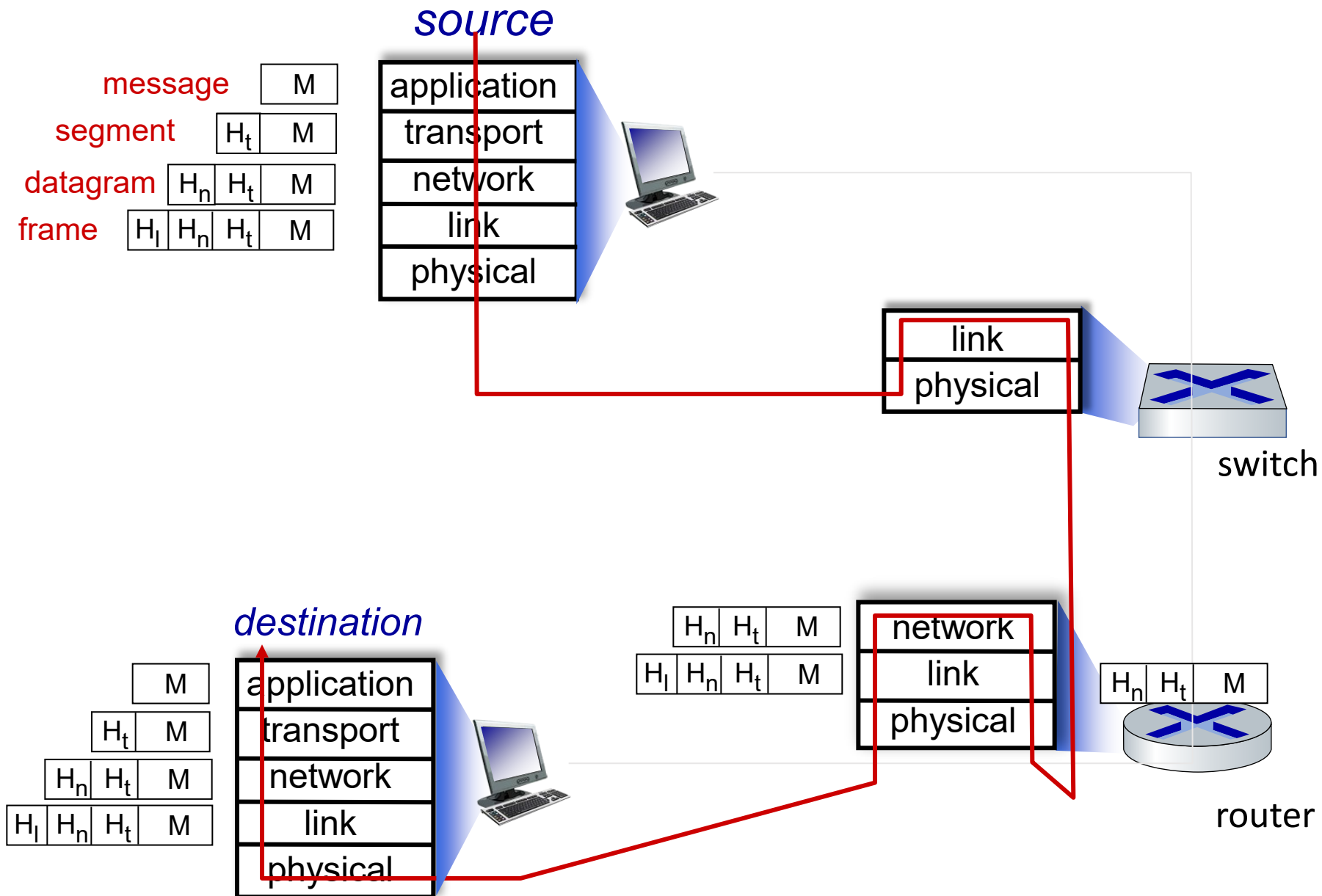
Matryoshka dolls (stacking dolls)



2.11 Services, Layering and Encapsulation



2.12 Encapsulation: an end-end view





2.13 Why Do We Use Layering?

- Layering reduces the complexity of a large system.
- Each layer provides a service to the layer above and uses services from the layer below.
- An implementation can change while the service interface remains compatible.
- If the physical medium changes from copper to fiber, must HTTP change?
 - No, the lower-layer implementation can change while higher-layer services remain usable.



Week 2 - Agenda

02

Protocol Layers and Service Models

03

Networks Under Attack

04

History of Networking and Internet

05

Next Week

3.1 Network Security

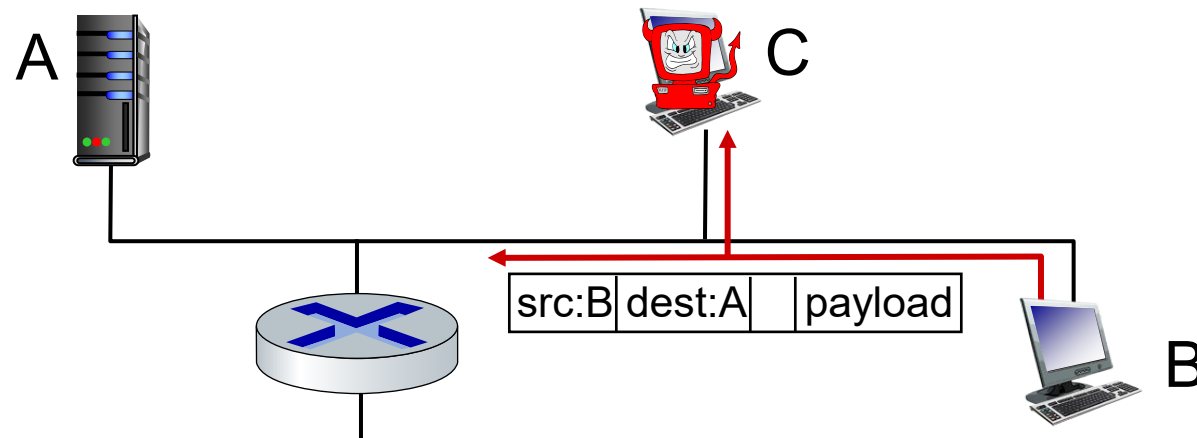
- Internet not originally designed with (much) security in mind
 - *original vision*: “a group of mutually trusting users attached to a transparent network” 😊
 - Internet protocol designers playing “catch-up”
 - security considerations in all layers!
- We now need to think about:
 - how bad guys can attack computer networks
 - how we can defend networks against attacks
 - how to design architectures that are immune to attacks



3.2 Packet Interception

packet “sniffing”:

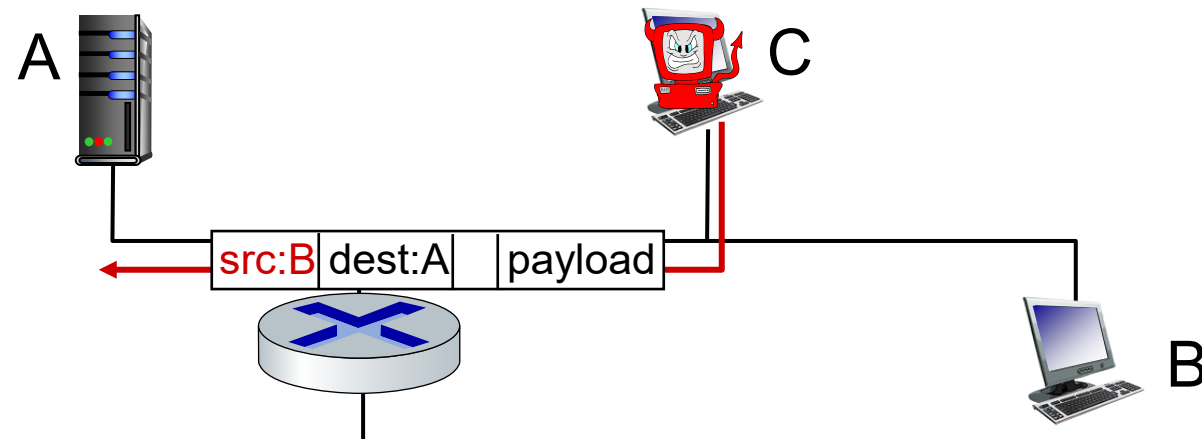
- broadcast media (shared Ethernet, wireless)
- Promiscuous (mode to listen all traffic) network interface reads/records all packets (e.g., including passwords!) passing by



Wireshark software used for our labs is a (free) packet-sniffer.

3.3 Fake Identity

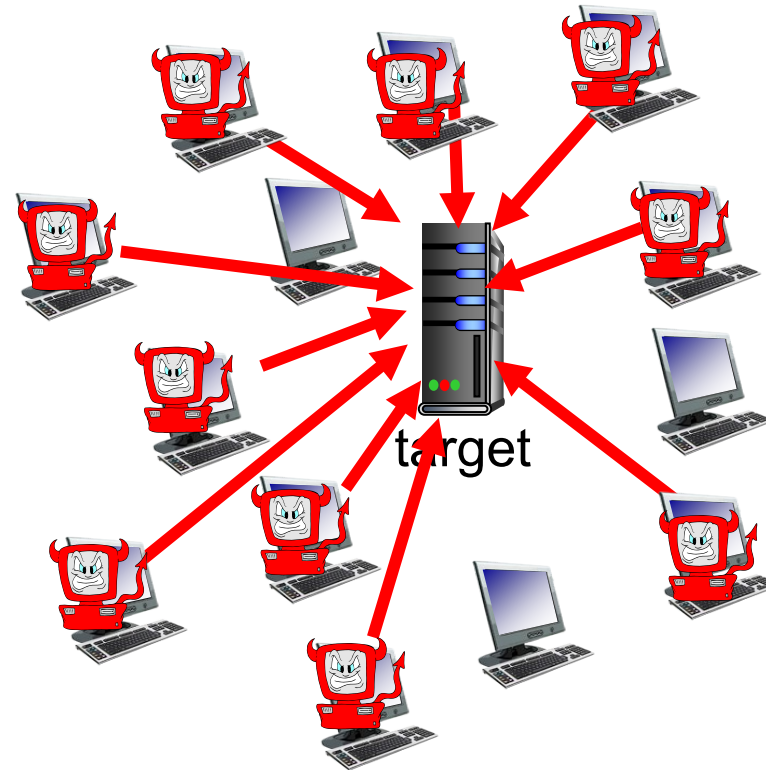
IP spoofing: injection of packet with false source address



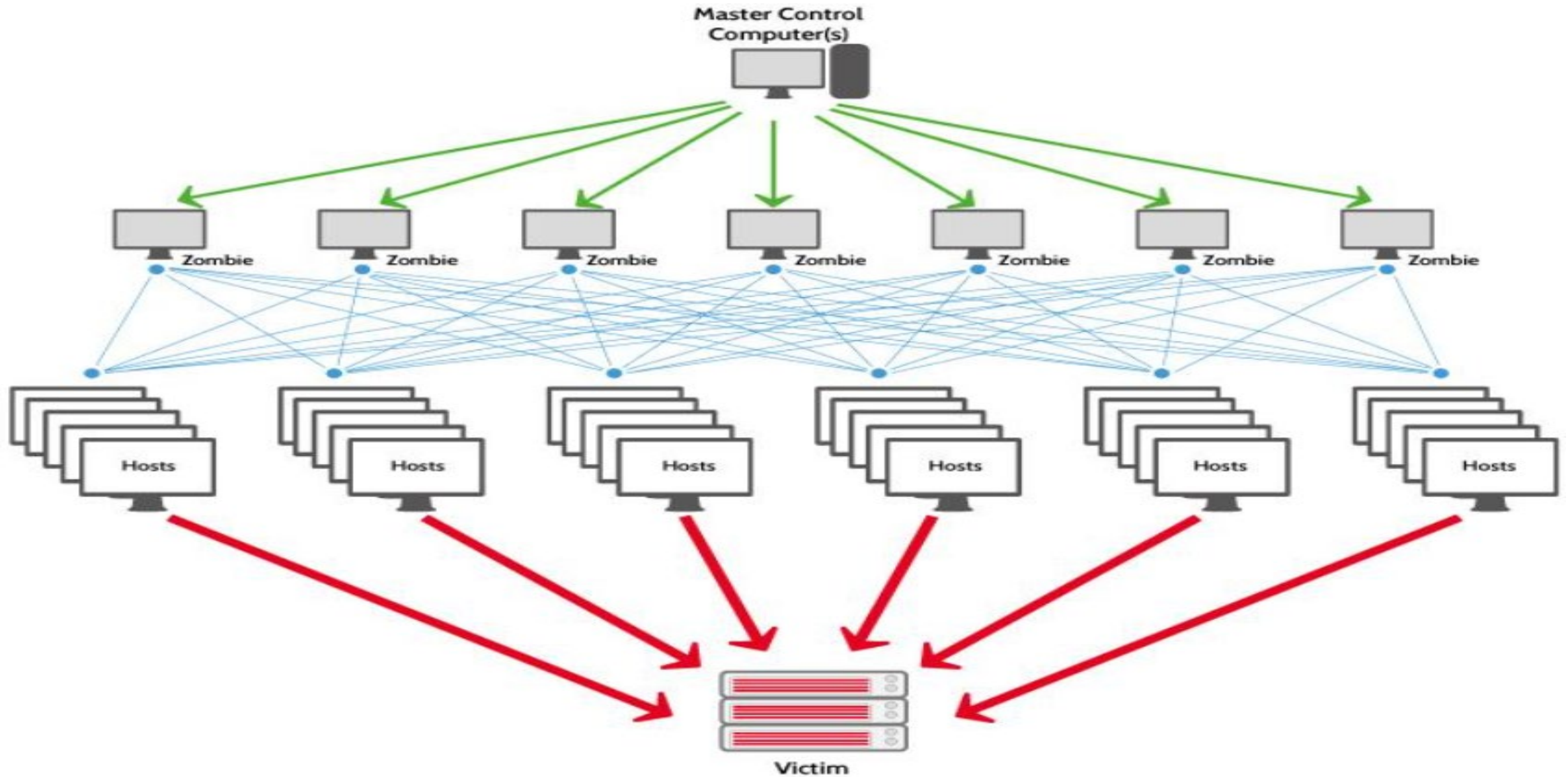
3.4 Denial of Service (DOS)

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate (tr: meşru) traffic by overwhelming resource with bogus (fake) traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



3.5 Denial of Service (cont.)





3.6 Security Concept Check

- Authentication: 'Who are you?'
 - Confidentiality: 'Can an unauthorized party read the content?'
 - Integrity: 'Was the content altered?'
 - Access control: 'Are you allowed to use this resource?'
 - Availability: 'Can legitimate users access the service?'
-
- These are different goals; one mechanism does not necessarily solve all of them.



3.7 Lines of Defense

- **Authentication**: proving you are who you say you are
 - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- **Confidentiality**: via encryption
- **Integrity Checks**: digital signatures prevent/detect tampering
- **Access Restrictions**: password-protected VPNs
- **Firewalls**: specialized “middleboxes” in access and core networks:
 - off-by-default: filter incoming packets to restrict senders, receivers, applications
 - detecting/reacting to DOS attacks



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4.1 1961-1972: Early Packet-Switching Principles

- **1961:** Kleinrock - queueing theory shows effectiveness of packet-switching
- **1964:** Baran - packet-switching in military nets
- **1967:** ARPAnet conceived by Advanced Research Projects Agency
- **1969:** first ARPAnet node operational
- **1972:**
 - ARPAnet public demo
 - NCP (Network Control Protocol) first host-host protocol
 - first e-mail program
 - ARPAnet has 15 nodes

1960s–1970s: packet-switching principles and ARPANET.

Why should we divide data into packets instead of sending it over a single continuous connection?



4.2 1972-1980: Internetworking...

- **1970:** ALOHAnet satellite network in Hawaii
- **1974:** Cerf and Kahn - architecture for interconnecting networks
- **1976:** Ethernet at Xerox PARC
- **late70's:** proprietary architectures: DECnet, SNA, XNA
- **1979:** ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

define today's Internet architecture

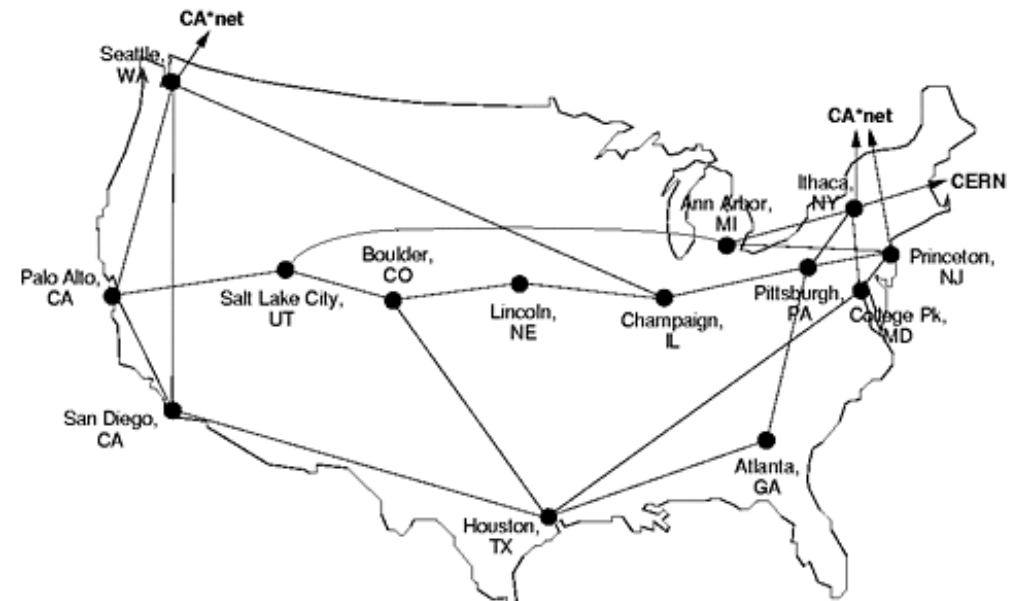
4.3 1980-1990: New protocols...

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: FTP protocol defined
- 1988: TCP congestion control

- new national networks: CSnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks

How can different networks communicate with each other?

NSFNET T1 Network 1991



4.4 1990, 2000s: Commercialization, the Web...

- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960's]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990s: commercialization of the Web

late 1990s – 2000s:

- more killer apps: instant messaging, P2P file sharing
- network security to forefront
- est. 50 million host, 100 million+ users
- backbone links running at Gbps

How did the Internet infrastructure evolve into an information environment accessible to users?



4.5 2005-present: Scale, SDN, Mobility, Cloud

- Aggressive deployment of broadband home access (10-100's Mbps)
- 2008: software-defined networking (SDN)
- Increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- Service providers (Google, FB, Microsoft) create their own networks
 - bypass commercial Internet to connect “close” to end user, providing “instantaneous” access to social media, search, video content, ...
- Enterprises run their services in “cloud” (e.g., Amazon Web Services, Microsoft Azure)
- Rise of smartphones: more mobile than fixed devices on Internet (2017)
- ~18B devices attached to Internet (2017)

Why is a modern web application not limited to a single computer?



Any Questions

5.1 Next Week

- We will focus on Application Layer.

Your duties:

- ✓ Download and Install WireShark application.
- ✓ Review this lecture note via learn.khas.edu.tr
- ✓ Read chapter 1&2 from Kurose & Ross's textbook.



Thank You
For Your Attention!

